

Abdelrahman Ahmed

Electrical Engineering Student

✉ bodebi639@gmail.com

🌐 <https://www.linkedin.com/in/bitboody/>

☎ +20 10 935 791

📍 Cairo, Egypt

WORK EXPERIENCE

Software Techies, Belgium (Remote)

Software Engineering Intern

Jun 2024 - Sep 2024

- Worked on optimizing a Node.js backend serving thousands of users, migrating it from JavaScript to TypeScript, and developing web scraping and data processing tools in Go to automate tasks and streamline workflows for engineers.

EDUCATION

MTI University

B.Sc Electrical Engineering

Sep 2024 - Present

PROJECTS

Odysseus (Open Source Maintainer)

- Serve as one of the maintainers of Odysseus, an open-source, self-hosted AI workspace platform with 86,000+ GitHub stars and a large developer community.
- Act as the core developer of the Tauri native port, contributing to its architecture and integration with the existing application.
- Develop features, resolve complex backend and application-level issues, and improve system reliability.
- Build CI/CD pipelines and developer tooling while reviewing pull requests, supporting contributors, and maintaining automated tests.

Radioactive decay simulator (Python)

- Developed a Python simulator for radioactive decay, calculating half-lives and predicting gamma-ray emissions.
- Implemented exponential decay models and visualized simulation results to explore nuclear physics concepts.

6502 CPU Emulator (C)

- Developed a cycle-accurate 6502 emulator in C, implementing the complete instruction set and addressing modes.
- Added support for interrupts and accurate processor status flag behavior.
- Modeled CPU execution with cycle-accurate timing.
- Verified emulator correctness against functional 6502 test ROMs.

Necromancer — Remote Administration PoC (Node.js)

- Developed a client-server remote administration proof of concept in Node.js for security research.
- Implemented TCP-based communication and remote command execution.
- Explored networking, process management, and distributed system concepts.